

A Benchmark for Graph-Based Dynamic Recommendation Systems

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Abstract

The surge of Graph Neural Networks (GNNs) has catalyzed significant advancements in recommendation systems by enabling more effective modeling of user-item interactions within undirected bipartite graphs. However, the proliferation of GNN architectures, coupled with the absence of standardized benchmarking frameworks, presents challenges in systematically evaluating and comparing different dynamic recommendation models. In response, we propose a comprehensive benchmarking study of bipartite GNN operators for recommendation systems using the PyTorch Geometric library. Our contributions include the development of a flexible benchmarking framework encompassing data preprocessing, model training, and evaluation protocols, facilitating fair comparison across diverse dynamic recommendation scenarios. We rigorously assess the performance of various GNN models, ranging from traditional methods to state-of-the-art architectures, on the MovieLens100k dataset. Through insightful analysis of experimental results, we elucidate the strengths and weaknesses of different GNN operators and offer practical suggestions for model selection and configuration. Our work aims to foster transparency, reproducibility, and innovation in GNN-based dynamic recommendation systems, providing a valuable resource for researchers and practitioners in the field.

Keywords: Recommendation Systems, Graph Neural Networks (GNN), PyTorch Geometric